Chapter 6

Occlusion

David Ricketts

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WHAT IS OCCLUSION?

Occlusion is the way in which the maxillary and mandibular teeth come together. This definition conjures up a static relationship; however, in function the teeth move across one another and this articulation or dynamic occlusion is equally important. These tooth contacts cannot be looked at in isolation as the masticatory system also involves the periodontium, the skeletal components (including the temporomandibular joints) and the neuromusculature.

For some the study of occlusion is shrouded in mysticism and for others there is a conviction in philosophy that has seen occlusal adjustments and rehabilitations carried out solely to ensure a patient's occlusion fits the perfect ideology. Such conviction has been based upon subjective workings of various individuals and little sound evidence. Every patient is an individual and their occlusal management should be customized accordingly. However, when extensive rehabilitation of the dentition is required, some of the traditional teachings are useful.

It is important when restorations are placed that they are in functional harmony with the masticatory system to ensure a comfortable functioning apparatus. Failure to do so has been implicated in the aetiology of temporomandibular disorders and bruxism.

BRUXISM

Bruxism is a parafunctional activity which involves the clenching and grinding of teeth. This can occur consciously when awake (awake bruxism) or at night when asleep

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(sleep bruxism). Awake bruxing is more common in females, has been linked to anxiety and stress, and is thought to affect 20% of the population. The prevalence of sleep bruxism decreases with age, with it being reported in 14–18% of children, 8% of adults and only 3% of elderly. Sleep bruxism is frequently noisy and reported by partners, and individuals may wake up with stiffness and aching of the jaws. Complaints of headaches are also common. Occlusal interferences were once thought to provoke sleep bruxism, but it is now thought of as a sleep-related movement disorder.

Most bruxism is mild and non-damaging to tooth tissue; as such, no treatment is required. However, for some, tooth wear and fracture may occur or the noise of grinding is unacceptable for partners and treatment is sought. For these patients a correctly adjusted hard acrylic splint can be supplied for night-time or daytime wear to protect the teeth. Such splints are referred to as stabilization splints and can be worn in the maxillary (Michigan splint, Figure 6.1; see also Chapter 4) or mandibular (Tanner appliance) arch.

An alternative splint is the localized occlusal interference splint (LOIS appliance, Figure 6.2). This consists of an acrylic plate retained by suitable clasps and two ball-ended wires which are placed between opposing teeth to interfere with the occlusion. Occlusal loading of the ball interferences leads to stimulation of periodontal mechanoreceptors, afferent feedback and reduction in occlusal loading. When used for bruxing habits, both splints act as a 'habit breaker' and patients can be gradually weaned off them.

If tooth destruction has taken place as a result of bruxism and restorative management is required, damage to any restorations can occur and it is important that the patient is aware of this. Damage to new restorations can be reduced by correct choice of dental materials, namely metal occlusal surfaces where possible and protection of restorations with a stabilization splint which can be worn at night for sleep bruxism and during the day if necessary for awake bruxism.

TEMPOROMANDIBULAR DISORDERS

Within the dental literature there seems to be no agreement as to a definition for temporomandibular disorders (TMD). What is clear is that TMD covers a number of

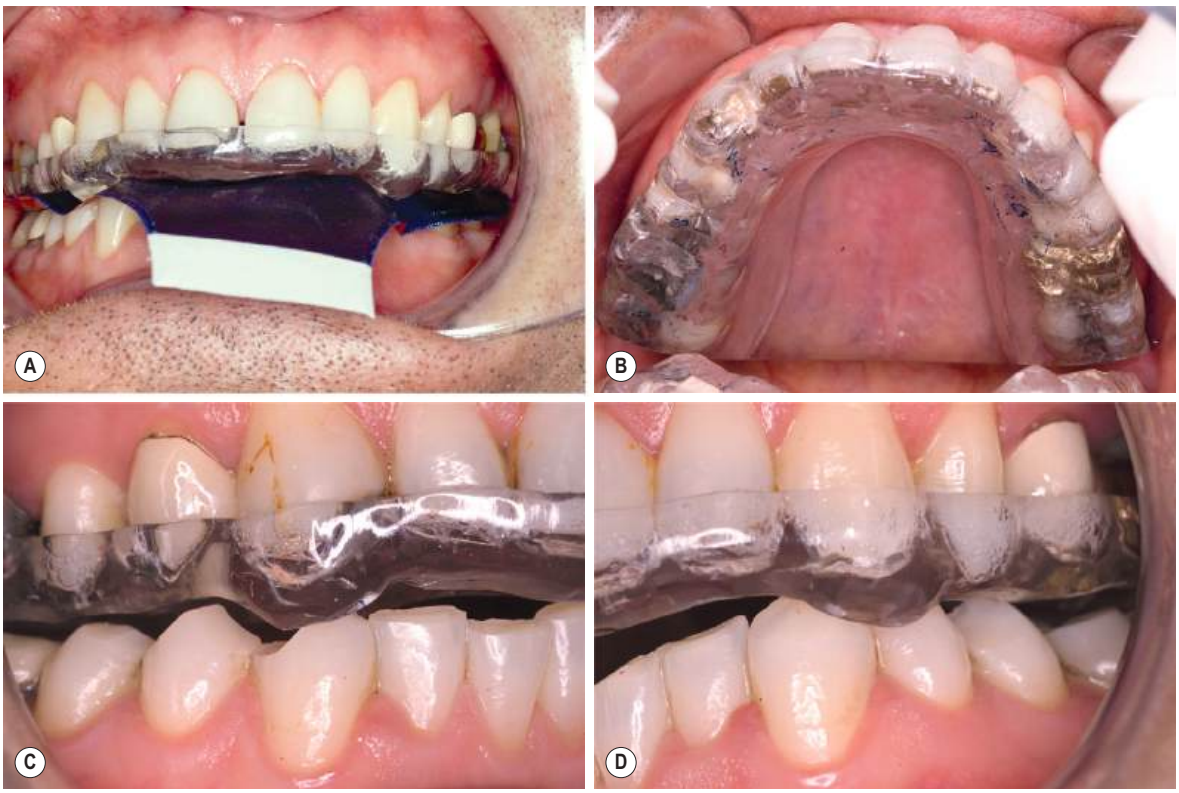


Figure 6.1 A Michigan splint being fitted and occlusion being checked to ensure all lower teeth occlude with the splint (A, B). Manufacture using casts mounted on an articulator reduce the amount of adjustment. The splint should have a canine rise in lateral excursions (C, D).

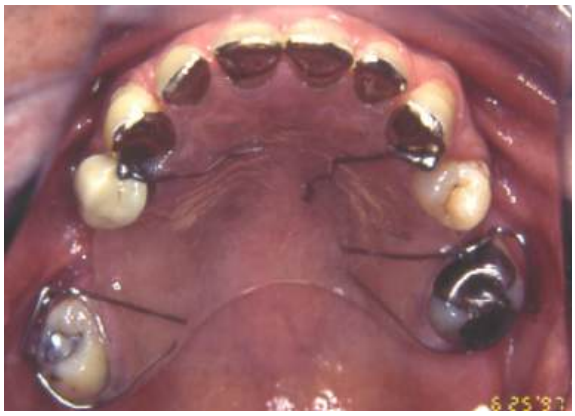


Figure 6.2 Localized occlusal interference splint (LOIS appliance). The ball-ended wires on the occlusal surface of the canine–premolar teeth interfere with the occlusion and are aimed at breaking a bruxism habit.
(Courtesy of Dr John Radford)

complex conditions with common signs and symptoms. The conditions are defined in the glossary of prosthodontics terms as those producing 'abnormal, incomplete or impaired function of the temporomandibular joint(s)'. The most common conditions making up these disorders, together with their signs and symptoms, can be seen in Figure 6.3. A number of terms have been given to the signs and symptoms allotted to pain dysfunction syndrome, some of which include myofascial pain, craniomandibular dysfunction and mandibular dysfunction. Rarer causes of TMD are rheumatoid arthritis, psoriatic arthritis, developmental defects, infection, neoplasia and ankylosis.

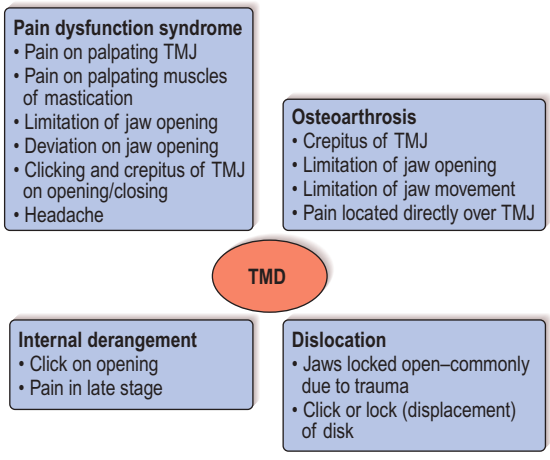


Figure 6.3 The most common conditions with their signs and symptoms which make up the temporomandibular disorders (TMD).

Epidemiological data indicate that signs and symptoms of TMD are relatively common (50–75% and 20–25%, respectively) and are evenly distributed between men and women; however, only 3–4% of the population actually seek treatment and these are more likely to be women, hence it is frequently and incorrectly cited that women are affected more than men.

Many theories on the aetiology of TMD have been put forward and, to a degree, management has been dictated by these thoughts. Whilst bruxism is a separate entity, it has been linked with TMD, possibly as one of the aetiological or exacerbating factors. Historically the occlusion (malocclusions, interferences and non-working side contacts) was also incriminated in causing TMD but as more is known about the condition it is realized that it has a complex and multifactorial origin and that occlusion plays a relatively minor role. Occlusal adjustments have been shown in numerous studies neither to prevent nor treat TMD and should therefore not be considered. TMD patients have been shown to have higher levels of stress, anxiety, depression and aggressive behaviour and as such it is thought that psychological factors may be more important.

TMD can be treated in a number of ways. Simple advice and reassurance that the condition is relatively common and not sinister may be sufficient. A simple change to a soft diet for a period of time, jaw rest and a hot pack over the tender areas can be advised. Medical management can include non-steroidal anti-inflammatory drugs which can be used topically (Mentholatum Deep Relief gel containing ibuprofen) or systemically if not contraindicated by any other medical condition such as asthma or gastritis. For the restorative dentist, the most common, time-honoured and least invasive form of treatment is to provide the patient with a stabilization splint. This treatment is often successful but why it is, is less clear. It may be: (1) a simple placebo effect; (2) the splint may disengage the occlusion, so eliminating any occlusal interferences and a reflex reduction in muscle activity; (3) the splint's repositioning effect may allow recapture of an anteriorly displaced disc; or (4) the increased occlusal vertical dimension alone may lead to relief. It has been shown that the greater the splint thickness, the more rapid the relief of symptoms. If these simple non-invasive treatments are not successful, thought should be given to referral to a specialist where other modalities of treatment such as prescription of muscle relaxants or antidepressants can be considered.

It is advisable to avoid complex and advanced restorative work in patients who are or have suffered from TMD. Their adaptation to the smallest changes in occlusion may be poorly tolerated and signs and symptoms of TMD may be exacerbated. Such patients can also be very focused on their 'bite' and have unrealistic expectations. If restorative treatment is unavoidable, treatment of the TMD needs to be carried out first and great care

needs to be taken to ensure that any changes to the occlusion are gradual and staged.

PATIENT EXAMINATION

To enable the correct management of the occlusion, an understanding of its components and their function is important; this unfortunately is frequently overlooked in a clinical examination. This section details what should be examined and recorded.

Extraoral examination

Facial appearance

The facial appearance can firstly be assessed for facial asymmetries and skeletal relationship. With the patient's teeth in contact, the lower face height should be assessed in proportion to the total face height, especially in patients complaining of missing teeth and tooth wear. Loss of teeth and occlusal stops could result in an over-closed appearance but a reduction in lower face height is unusual in patients with tooth wear. This is because of dento-alveolar development or compensation which counterbalances the loss in coronal tooth tissue height. In fact it has been suggested that dento-alveolar development takes place in the absence of tooth wear and some researchers have shown an increase in face height with age. This is the mechanism which leads to overeruption of teeth when taken out of occlusion – for example, by loss of an opposing tooth (Figure 6.4).

Temporomandibular joints (TMJ)

The patient should be asked to open and close their mouth. The extent to which they can open should be measured between the incisal edges of the upper and lower incisor teeth. At maximum opening this should be more than 35 mm for women and 40 mm for men. Any limitation to opening should be recorded. Any deviation of the mandible on opening and closing from a normal vertical straight line should be observed and described. Gentle pressure over the head of the condyle should be applied when the patient opens and closes. This should be repeated with a finger in the external auditory meatus. Any tenderness, clicking or crepitus (grating sensation) should be noted and when they occur in the opening–closing cycle.

Intraoral examination – examination of the occlusion

Intercuspal position (ICP)

Intercuspal position can be defined as the position of the jaws when the maxillary and mandibular teeth are in maximum intercuspation. This has also been referred to as centric occlusion. Patients usually close from a rest position immediately into this position due to a conditioned path of closure. It is important to establish whether the occlusion in ICP is stable; that is, an occlusion where there is no possibility of tooth movement, namely overeruption, drifting or tilting of teeth (Figure 6.5). If teeth are extracted or lost due to other causes, or if they lose their contour (a carious lesion cavitating, tooth fracture, tooth wear), occlusal contacts and interdental contacts, teeth can move, often complicating any subsequent restorative work (Figures 6.4 and 6.6).

When teeth overerupt in the absence of periodontal disease, the alveolar process remodels and the gingival margin moves with the tooth (see Figure 5.3, Chapter 5). This dento-alveolar compensation can happen quickly in some patients and is a reason why temporary crowns



Figure 6.4 Loss of the lower left first and second molar teeth and upper left first and second premolar teeth has allowed overeruption of the opposing teeth. This now complicates their replacement as there is reduced height above the edentulous ridges.



Figure 6.5 Patient with a stable intercuspal position (ICP): all interdental and occlusal contacts are intact preventing any tooth movement.



Figure 6.6 Patient with a heavily restored dentition. Repeated loss of the restoration in the lower left first premolar tooth has allowed overeruption of the upper left canine.
(Courtesy of Suzanne Blacker)

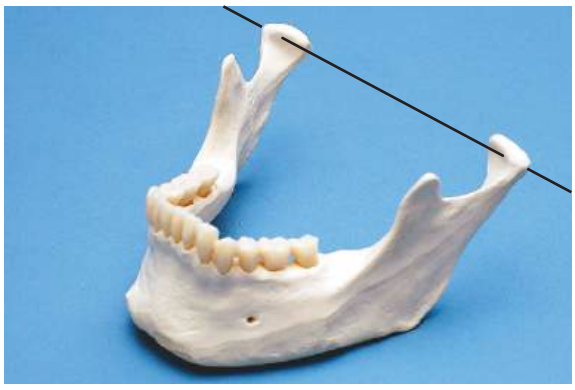


Figure 6.7 When the mandible is bilaterally in the most comfortable posterior location in the glenoid fossa, the mandible opens and closes around an arc of curvature through an imaginary axis drawn through the centre of the head of both condyles. This imaginary axis is termed the terminal hinge axis.

should be placed following tooth preparation for crowns or bridges; the occlusal and proximal reduction will result in an unstable occlusion and if no temporary is placed, overeruption and drifting of teeth could occur. This would result in a prosthesis that does not seat fully and appear to be occluding prematurely or appear 'high' in ICP.

Centric relation, terminal hinge axis position and retruded contact position (RCP)

Centric relation describes the jaw relationship between the maxilla and mandible when the mandible is in a retruded position. Differing definitions of centric relation have focused on slightly different positions of the condyle in the glenoid fossa and some on the relationship of the head of the condyle to the interarticular disc. These are rather academic arguments as clinically the position of the condyle cannot be visualized or confirmed without complex equipment. A more pragmatic and practical definition was proposed by Christensen in 2004 which conforms to most dentists' clinical practice. He described centric relation as the 'most comfortable posterior location of the mandible when it is bilaterally manipulated gently backward and upward into a retrusive position'. When this is done the mandible opens and closes on an arc of curvature around an imaginary axis drawn through the centre of the head of both condyles; this imaginary axis is termed the *terminal hinge axis* (Figure 6.7). Measured in the incisor region the arc of opening around the terminal hinge axis position takes place for about 20 mm before the condyles start to translate down onto the articular eminence (Figure 6.8). When the mandible closes in the terminal hinge axis position the first tooth contact is called the *retruded contact position* (*RCP*). The terminal hinge axis position is said to be the most reproducible jaw relationship; however, small variations from day to day and at different times during the day may occur. For most dentate patients there is a slide of the teeth and mandible from RCP into ICP of about 1–2 mm in an anterior and upwards direction. It is important to identify the RCP contact and record the direction and smoothness of the slide.

Excursive movements of the mandible

Retruded contact position is the most posterior tooth contact. All movements of the mandible occur anterior to this and can be guided by either the teeth when in contact and limited by the musculoskeletal system when the teeth are together or on jaw opening. With the teeth together the mandible can move forwards (protrusion), directly to the left or right (left or right lateral excursions) or in a pathway between these extremes. Some texts refer to anterior and posterior guidance, where the anterior guidance is provided by the tooth contacts and posterior guidance by the temporomandibular joints and associated ligaments, disc and musculature. In this terminology, the anterior guidance does not necessarily have to be on the anterior teeth; it can be on posterior teeth.

Posterior guidance

If the TMJ is viewed in three dimensions, its movements can be described when the mandible is moved into protrusion and lateral excursion.

Protrusion

When the mandible moves into protrusion, the condyles move from the glenoid fossa in a forward and downward movement onto the articular eminence. The angle this path makes with the horizontal when the individual is sitting upright is termed the *condylar inclination* (Figure 6.9).

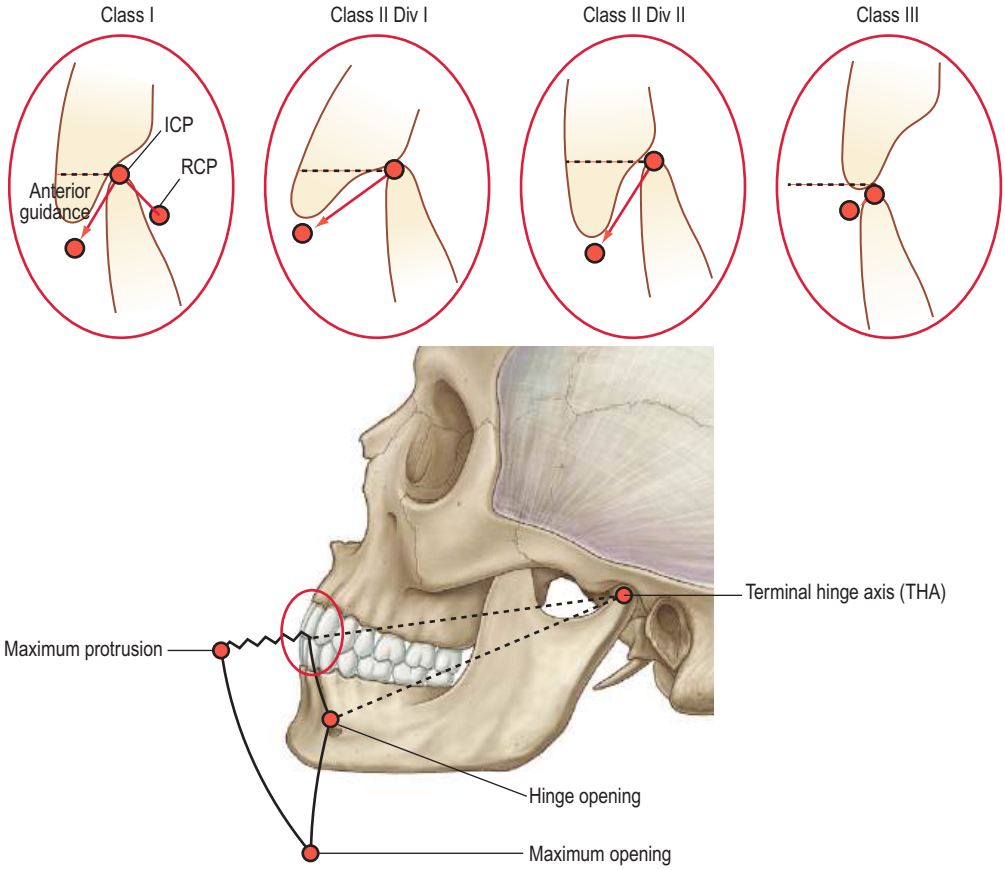


Figure 6.8 Main picture depicts Posselt's envelope of movement – a tracing made by an imaginary point between the lower incisor teeth in the sagittal plane. The inset images show how the incisor relationship can affect the amount of anterior guidance. ICP, intercuspal position; RCP, retruded contact position.

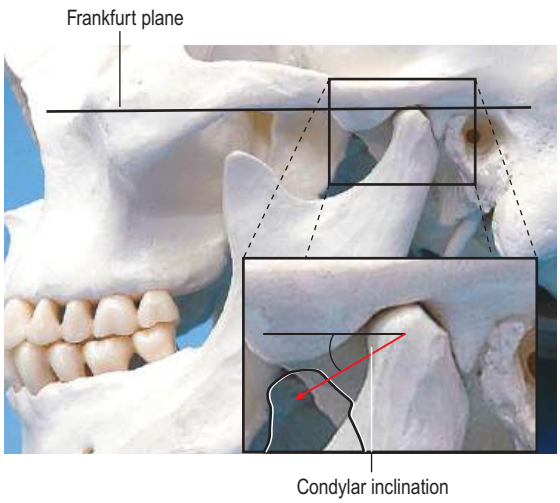


Figure 6.9 In protrusion the condyle moves down and forward onto the articular eminence. The angle this makes with the horizontal is the condylar guidance.

Lateral excursion

In lateral excursion, the side to which the mandible moves is called the working side and the opposite side from which the mandible moves is the non-working side. In Figure 6.10 the inferior surface of the skull can be seen with the outline of the mandible superimposed; the mandible is moving to the right and hence the right-hand side is the working side and the left-hand side the non-working side.

When the mandible moves into lateral excursion the non-working condyle moves forward and medially; the angle this makes with the parasagittal plane is termed Bennett angle (Figure 6.10) and is viewed in the horizontal plane. The working side condyle also moves laterally and this is called Bennett movement.

Anterior guidance

Protrusion

In protrusion the posterior teeth or anterior teeth may contact and this is governed by the incisor relationship

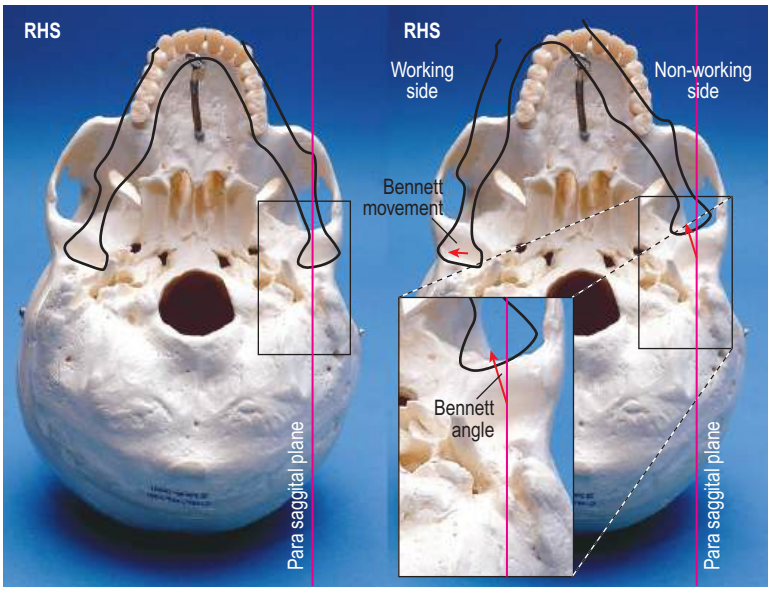


Figure 6.10 View of base of skull with outline of mandible in intercuspal position superimposed (left). In right lateral excursion (right) Bennett movement and Bennett angle can be seen.

and incisal guidance. There are four classifications describing the incisor relationships based upon their inclination, and horizontal (overjet) and vertical (overbite) overlap (see Figure 6.8):

- *Class I*. The lower incisors occlude within the middle third of the palatal surface of the upper incisors (Figure 6.11). In Class I incisor relationships the overjet is normally 2–3 mm and the overbite 2–4 mm. As the mandible moves forwards the mandible drops and the posterior teeth usually separate or disclude.
- *Class II*. In a Class II incisor relationship the lower incisors occlude or potentially occlude with the gingival third of the palatal surface of the upper incisors or palatal gingivae. This classification is divided into two subdivisions depending on the inclination of the upper incisors:
  - Class II Division I. The upper incisors are proclined and the overjet is increased. Depending on the severity of the incisor relationship, the incisal guidance is reduced and the likelihood of posterior teeth contacting in protrusion is increased.
  - Class II Division II. The upper incisors are upright or retroclined and the overbite is increased. In protrusion the incisal guidance is marked and the posterior teeth separate or disclude immediately (Figure 6.12). Posterior tooth contacts are therefore unlikely.
- *Class III*. In Class III incisor relationships the lower incisors occlude with the incisal third of the upper incisor teeth, the teeth may have an edge-to-edge relationship or the lower incisors may occlude in front of the upper incisors (Figure 6.13), a so-called reverse overjet. In this incisor relationship the incisal guidance is reduced and the posterior teeth are likely to contact in all excursive movements.



Figure 6.11 Patient with a Class I incisor relationship.

In addition to these incisor relationships, there may be an anterior open bite (Figure 6.14), where there is no anterior tooth-to-tooth contact or there is bimaxillary proclination where both upper and lower anterior teeth are proclined. In both these situations incisal guidance is non-existent or reduced, respectively.



Figure 6.12 Patient with a Class II Division II incisor relationship. The deep overbite means that in protrusion the posterior teeth immediately disclude, making the placement of posterior restorations much easier and anterior restorations much more difficult.



Figure 6.13 Severe Class III incisor relationship with reverse overjet. In protrusion there is no guidance on the anterior teeth and posterior teeth will remain in contact, making the occlusion on posterior restorations more difficult to achieve.



Figure 6.14 Patient with a severe Class II Division I incisor relationship and an anterior open bite. In all excursive movements the posterior teeth are in contact and making posterior restorations will be more difficult.



Figure 6.15 In lateral excursion the guidance on the working side can either be canine guided (top) or group function (bottom).

Lateral excursion

In lateral excursion tooth contacts on the working side can be either between the canine teeth, termed canine guidance, or between one or more posterior teeth, termed group function (Figure 6.15).

Whilst guidance in lateral excursion is usually provided by the teeth on the working side, non-working side contacts can occur (Figure 6.16). This is more likely if the condylar inclination is shallow or if the tooth guidance on the working side is shallow. Both will mean that the body of the mandible moves more horizontally into lateral excursion with the non-working side teeth not dropping downwards and hence more likely to make contact.

TWO APPROACHES TO MANAGING THE OCCLUSION

Current management of the occlusion in operative dentistry is very much based on what is regarded as good clinical practice and will depend upon whether single



Figure 6.16 This patient is moving into left lateral excursion. There is an obvious non-working side contact between the molar teeth.

individual restorations are placed or whether extensive work means that all or most of the occlusal surfaces are being restored. In the former situation when individual or small numbers of restorations are being placed a *conformative approach* to the occlusion is adopted whereby the restoration(s) is made to fit in harmoniously with the existing occlusion and articulation. This does not mean that the occlusion is adjusted until no 'high spots' are detected, but rather simultaneous, even occlusal contacts need to be achieved with both the restoration placed and the adjacent teeth. When most or all of the teeth are to be restored with fixed prosthodontics or a combination of fixed and removable prostheses, it is usual to adopt a *reorganized approach* to the management of the occlusion. Here the occlusion is reorganized so that the RCP and ICP coincide at a predetermined vertical dimension.

ARTICULATORS

To assist in the detailed examination of the occlusion, aid treatment planning and facilitate the laboratory in making indirect restorations, study casts are often mounted on an articulator. An articulator is a hinged device connecting maxillary and mandibular members, onto which corresponding casts can be attached, enabling the relationship between opposing teeth to be reproduced as accurately as possible. Articulators vary in their complexity and hence in their ability and accuracy to reproduce jaw movements.

Probably the most commonly used method to locate opposing casts is to hand-hold them. This simple but effective method is only suitable for individual crowns in a Class I incisor relationship. In accepting hand-held models the dentist should appreciate that some intraoral occlusal adjustment will be needed at the chairside. If the occlusion is group function or the tooth involved with guidance, the time needed for occlusal adjustment can be excessive. Articulators attempt to mimic the relationship between the upper and lower jaws and should reduce the time needed to adjust an indirect restoration to fit the existing occlusion.

Simple hinge articulator

In its simplest form there is a simple hinge articulator (Figure 6.17), which is often used for simple fixed prosthodontics as only opening and closing of the jaws is possible. The arc of opening on the simple hinge articulator is different from that of a patient as the hinge is closer to the teeth on the model than the TMJ would be to the teeth in the mouth. Lateral excursive and protrusive movements are also not possible.

Average value articulators

Average value articulators allow for some degree of protrusive and lateral excursive movements (Figure 6.17). The condylar inclination is set at a single predetermined angle by the manufacturer which is usually 30° and not adjustable. The Bennett angle and Bennett movement are not reproduced and the maxillary cast is usually mounted in an arbitrary position. Whilst average value articulators allow some movement of the casts, this is not precise and therefore their use is limited to the reproduction of intercuspal position. They are not commonly used as they are neither semi-adjustable articulators which give more precision, nor do have they any significant advantage over hand-held models or simple hinge articulators.



Figure 6.17 A simple hinge articulator (left) and an average value articulator (right). Two further average value articulators are seen inset.

Semi-adjustable articulators

When planning and executing more advanced restorative dentistry, the semi-adjustable articulator is invaluable (Figure 6.18). The design of articulators varies with manufacturers but most allow adjustment of condylar inclination, Bennett movement and Bennett angle. In relation to articulators, the Bennett movement is often referred to as immediate side shift and the Bennett angle as progressive side shift. The height of the incisal pin can also be altered and the incisal table can usually be adjusted or modified/customized. The intercondylar width cannot be altered and is usually set at 11 cm.

Arcon vs non-arcon articulators

Semi-adjustable articulators can be either arcon (*ar*ticulating *con*dyle) or non-arcon in design. An arcon articulator is one in which the condylar component or sphere is attached to the mandibular member of the articulator and the glenoid fossa component to the maxillary member; it is therefore anatomically correct and thus easier to conceptualize. A non-arcon articulator is one in which the condylar component or sphere is attached to the maxillary member and sits in a track connected to the mandibular member.

Fully adjustable articulators

Fully adjustable articulators are more complex devices allowing the clinical scenario to be most closely reproduced. Instead of flat tracks and planes that reproduce the condylar movements on semi-adjustable articulators, fully adjustable articulators have further components that can be adjusted and use curved condylar inserts that can more accurately reproduce the three-dimensional nature of the glenoid fossa anatomy. These articulators require more complex information and time to program them, and pantographic (see Cadiax system) and stereographic recordings are needed. As these articulators are only as accurate as the recordings used to program them and are usually reserved for the most complex of restorative procedures, the semi-adjustable articulator is the articulator of choice for the vast majority of clinical situations.



Figure 6.18 An arcon (Denar, Whip Mix Co.) semi-adjustable articulator (left) and a non-arcon (Dentatus, Dentatus Company) semi-adjustable articulator (right).

Mounting models on a semi-adjustable articulator

A facebow and interocclusal record are required to mount maxillary and mandibular casts onto an articulator.

Facebows

Facebows are calliper-like devices that record the relationship of the maxillary teeth to the position of the condyle when it is in the terminal hinge axis position and allows this information to be transferred to the articulator when mounting dental casts. To use a facebow, firstly the position of the condyle needs to be identified and this can only be done accurately with a kinematic facebow (Figure 6.19). This consists of a clutch, which is essentially an impression tray that is fitted to the mandibular teeth, and a rigidly attached facebow which has condylar pointers on arms which can be extended or retracted and an adjustable angulation mechanism. The patient is then guided into the terminal hinge axis position and asked to open and close slightly. The adjustable components on the facebow arms are then altered until the condylar pointers no longer arc and pure rotation occurs: the condylar pointers are then directly over the terminal hinge axis.

Identifying the hinge axis position using a kinematic facebow is time consuming and most clinicians use arbitrary positions. The use of an earbow is convenient and



Figure 6.19 Diagrammatic representation of a kinematic facebow.

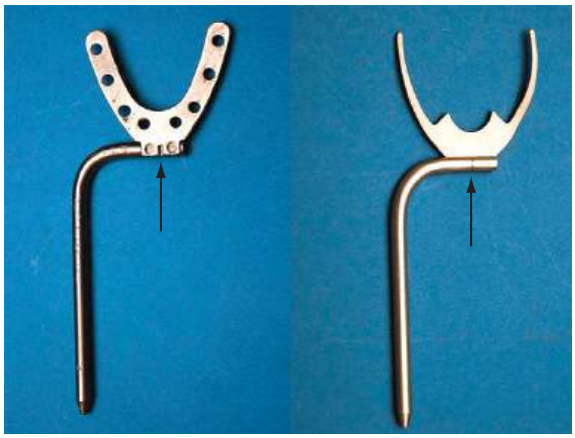


Figure 6.20 Bite fork for Denar and Dentatus facebows. Note both have a notch on the upper surface (arrowed) to allow alignment with the patient's centre line. The bite fork can be covered with either wax or an addition cured silicone bite registration paste.

in nearly 90% of patients the earbow axis will be within 6 mm of the true terminal hinge axis. The other most commonly used arbitrary hinge axis position is 13 mm along an imaginary line from the tip of the tragus of the ear to the outer canthus of the eye.

Once the positions of the condyles have been located, an imprint of the cusp tips and incisal edges of the teeth in the maxillary arch need to be recorded on a bite fork (Figure 6.20). Dental wax such as Beauty Pink wax or a silicone bite registration paste is an ideal material for this. It is important that the bite fork is seated firmly on the teeth and not rocked in an attempt to get an imprint of all teeth whilst the wax is cooling or when the silicone is setting. This ensures that the dental cast made in the laboratory seats precisely into the imprint without rocking during its mounting on an articulator. The bite fork is then linked to the facebow via a jig. The bite fork is then reseated on the maxillary teeth and held firmly in place, ideally by the patient. The condylar locators are then placed into position either in the external auditory meatus in the case of the earbow, or over the marked arbitrary or kinematically determined hinge axis position. The facebow needs to be centralized around the patient's head, which in the Denar Slidematic earbow is automatic and in the Dentatus facebow is achieved by ensuring the measurements on two rulers attached to each condylar pointer are equal.

To allow the occlusal plane of the maxillary cast to be set on the articulator to the correct angulation to the horizontal or Frankfurt plane (plane between the inferior border of the orbit or *orbitale* and the superior margin of the corresponding external auditory meatus or *porion*) in a patient standing upright, a third reference point is needed. The third reference point chosen will depend upon the facebow and articulator system being used; for example, the Denar MK II (Whip Mix Co.) uses a point on the face which is 43 mm above the incisal edge of the lateral incisor and the Dentatus (Dentatus Company) uses the *orbitale*. Finally, a pointer is aligned with the third reference point and all components tightened together.

The entire facebow (Dentatus) or jig and bite fork (Denar) can then be released from the patient's head, disinfected and sent to the laboratory for mounting of the maxillary cast. Figures 6.21 and 6.22 show how the Denar facebow is recorded and transferred to the laboratory for mounting of maxillary casts on the Denar MK II articulator, and Figures 6.23 and 6.24 show how this is done for the Dentatus facebow and articulator.

Interocclusal record

To enable the technician to mount the mandibular cast on the articulator in the correct relationship to the maxillary cast, an occlusal record is required. If all movements of the mandible are to be reproduced, including the RCP–ICP slide, a terminal hinge axis record should be taken; however, if the articulated casts are to be used to aid the manufacture of simple restorations, an ICP record is frequently sufficient.

Taking a terminal hinge axis record can be difficult in some patients due to preconditioned paths of closure as a result of feedback from periodontal mechanoreceptors and proprioceptors during repeated tooth contacts. Muscle tension or muscle splinting may also make it difficult to guide the mandible back into a posterior position. Even once an RCP contact has been made, for many patients the tooth contact leads them to readily slide into ICP. In these patients a Lucia jig or deprogramming appliance can be constructed at the chairside (Figure 6.25A). This is essentially a cold cure acrylic anterior bite platform which covers the upper central incisors. It is designed so that the lower incisors occlude at 90° to the platform with all posterior teeth separated. Repeated contact on the splint breaks down the programmed paths of closure and allows the operator to guide the mandible into its posterior position. The terminal hinge axis interocclusal record can be taken with the Lucia jig in situ (Figure 6.25B), thus preventing any sliding into ICP. Once mounted on the articulator, the incisal pin can be moved until RCP contact is made; this is possible due to the pure rotational closure of the mandible in the terminal hinge axis position. Other methods for deprogramming have been described in order to make a terminal hinge axis record and these include the use of cotton wool rolls between the teeth and a hard occlusal splint.

If sufficient teeth contact, an ICP record or terminal hinge axis interocclusal record can be taken in either

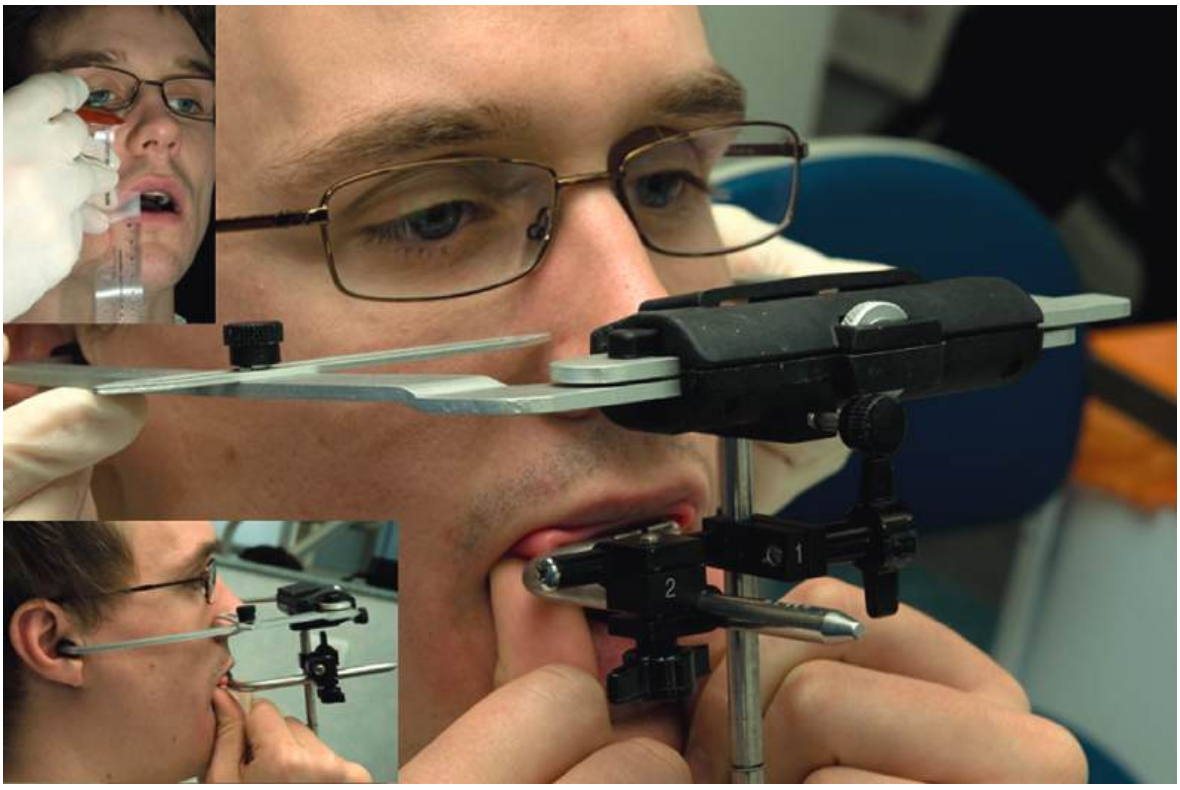


Figure 6.21 Denar Slidematic facebow (earbow) being recorded. The third reference point is recorded first and marked on the side of the nose with a pen (inset, top left). The condylar locators are placed in the external auditory meatus (inset, bottom left) by sliding the bow together. The bite fork and the slidematic earbow are then linked together with the jig making sure the numbers 1 and 2 on the jig face forwards. Once the jig has been adjusted so that the third reference pointer is aligned with the marked third reference point, the screws on the jig can be tightened.

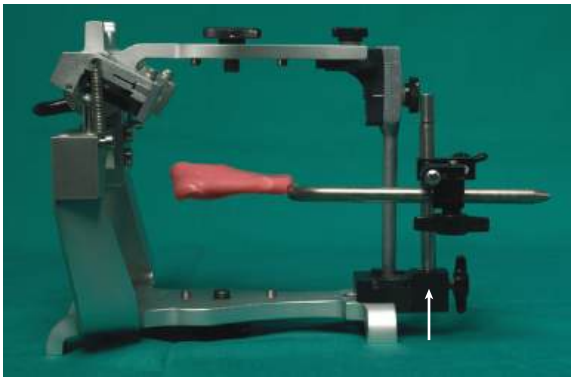


Figure 6.22 Denar bite fork and jig transferred to the Denar articulator. The pillar of the jig sits in a special mounting block (arrowed). The maxillary study cast can now be seated into the wax imprint on the upper surface of the fork and attached to the maxillary arm of the articulator with anti-expansion plaster.

wax or a silicone bite registration paste. If wax is used it is important not to use a 'horseshoe'-shaped record as this can easily distort in transit to the laboratory, and as it is not elastomeric, its original shape is not reformed. Instead the wax should be at least double thickness and extend across the arch. Once the patient has closed down into the warmed and softened wax, it can be removed and the excess cut away with scissors so that the technician can confirm that the study casts are fully seated into the record. This will obviously distort the wax and so it has to be reseated in the same position in the patient's mouth and the record re-established. Once removed from the mouth it is chilled in cold water and disinfected ready to send to the laboratory (Figure 6.26).

Silicone bite registration pastes are presented in double cartridges containing the base and catalyst pastes. These are inserted into a gun and syringed around the occlusal plane of the lower teeth through a double helix mixer tip which ensures a bubble-free, perfectly mixed paste

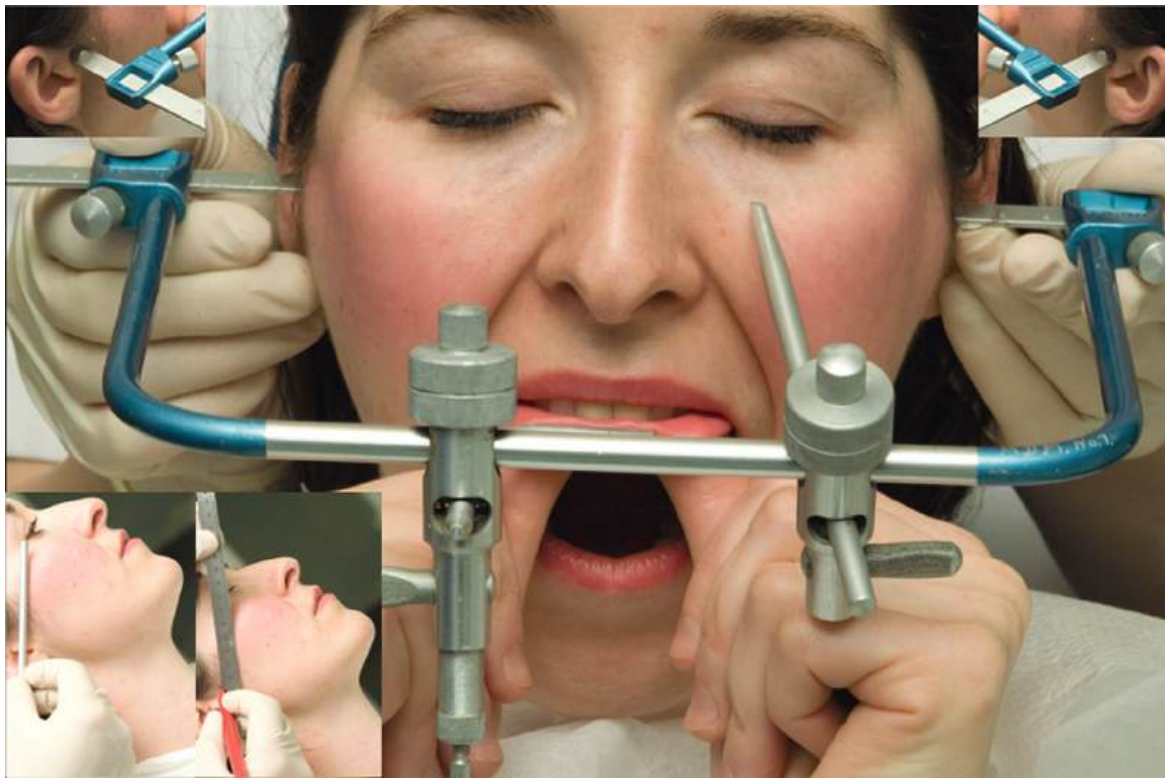


Figure 6.23 Dentatus facebow being recorded. The condylar pointers are placed over the marked arbitrary hinge axis position (inset, bottom left). The facebow is centralized by ensuring that the condylar rulers read the same on left and right hand side (inset, top). The orbital pointer is placed in position and the screws tightened.



Figure 6.24 The entire Dentatus facebow recording is then transferred to the Dentatus articulator. The pin at the foot of the bite fork jig (arrowed white) is adjusted until the orbital pointer touches the orbital plane on the articulator (arrowed black). The maxillary study cast is then placed on the bite fork imprint and the cast attached to the maxillary arm of the articulator using anti-expansion plaster.



Figure 6.25A Anterior and palatal view of Lucia jig made from cold cure acrylic.

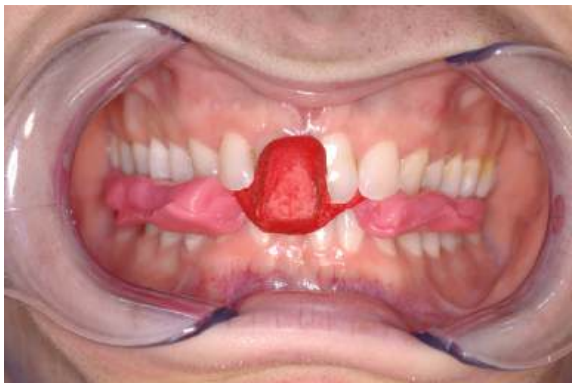


Figure 6.25B Wax terminal hinge axis interocclusal record being taken with the Lucia jig in situ.

(Figure 6.27). Bite registration pastes have a rapid set, avoiding patient fatigue when keeping the jaw in the same position and errors due to jaw movement.

Placing any material between the teeth and then the casts in the laboratory always has the potential to introduce errors; therefore, if the occlusion is obvious, the casts can be manually placed together using marked index teeth (Figure 6.28). If insufficient teeth are present, wax occlusal record blocks may be required and used as for recording the occlusion for removable prostheses.

Programming the semi-adjustable articulator

The adjustable components on a semi-adjustable articulator are commonly set at average values: condylar inclination 25–35°, progressive side shift (Bennett angle) at 7–15° and immediate side shift (Bennett movement) 1–2 mm.

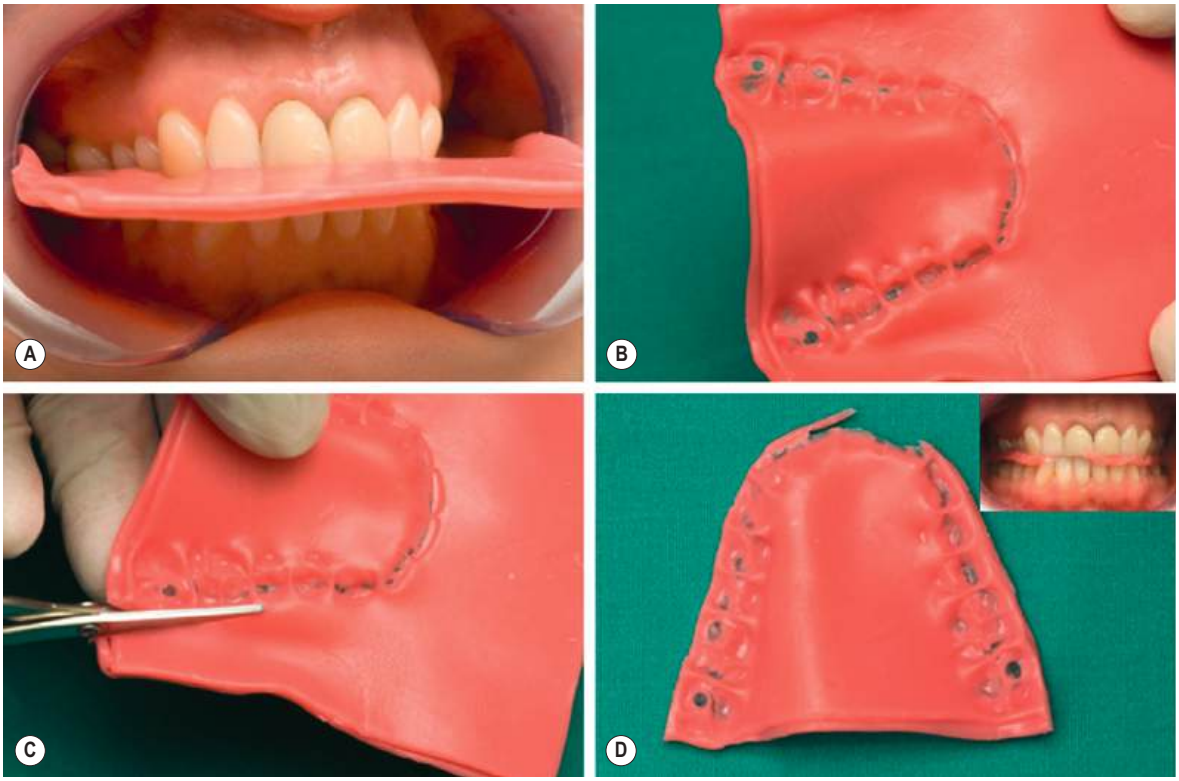


Figure 6.26 Steps for taking a wax interocclusal record. The patient bites into at least a double layer of warmed and softened wax, ideally wax such as Beauty Pink wax (Moyco Union Broach, York, PA) which is hard and brittle when cold (A). When removed (B) and still warm and soft, the excess wax is removed using a scissors (C). As this distorts the wax it is reseated in the mouth, removed again and chilled, disinfected and sent to the laboratory (D).

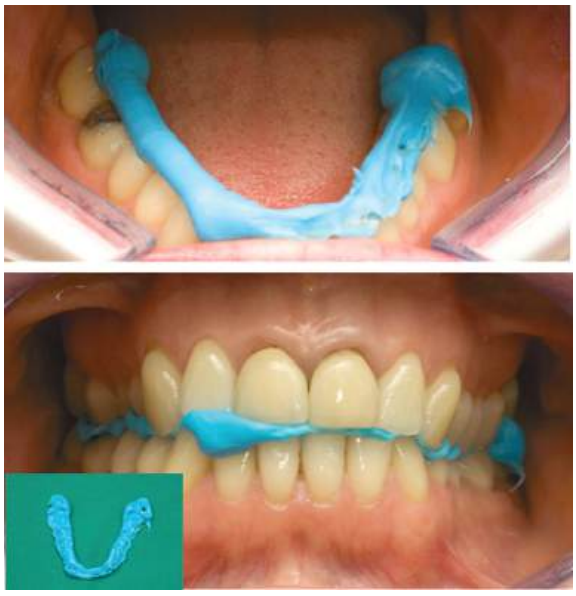


Figure 6.27 Silicone bite registration paste syringed over the occlusal surfaces of mandibular teeth (top). The patient bites through the paste (bottom) and once set is removed, disinfected and sent to the laboratory (inset).

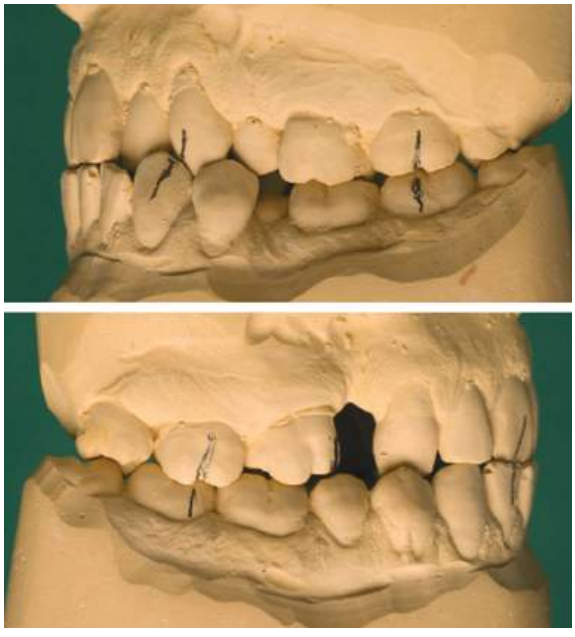


Figure 6.28 Study casts with index teeth marked. The patient's occlusion is obvious and no interocclusal record is needed.

To customize the settings for each patient, excursive occlusal records are required or a pantographic tracing. This is described here using the Denar MK II articulator as an example. The operator should start with all settings on the Denar MK II articulator set at zero. To set the condylar inclination a protrusive wax record is required. Protrusion of the mandible brings the condyles down onto the articular eminence and the posterior teeth will usually separate. This will also depend on the amount of incisal guidance. A protrusive wax record will record this amount of separation. The maxillary and mandibular arms of the articulator should be separated and the protrusive record placed between the casts until they are fully seated. This will bring the condylar spheres downward and forward as in the patient. The condylar inclination element is then released and tilted down until it contacts the condylar sphere and secured; this is the condylar inclination.

To set the Bennett angle and Bennett movement, left and right lateral excursive records are required. In left lateral excursion the left condyle moves laterally and the immediate side shift adjustment is made on the left glenoid fossa component of the articulator until the medial wall contacts the condylar sphere. On the non-working side (right-hand side) the condyle would have moved forward and medially and the Bennett angle component is again adjusted until the condylar sphere seats into the fossa and against the medial wall. The same is repeated with a right lateral excursive record. Figure 6.29 shows the adjustable components on the Denar MK II articulator. Some dentists simply use protrusive records only to set condylar inclination and then set Bennett angle and movement to average settings, and some take lateral records only to set all components.

Alternatively, pantographic tracings or readings can be used. A pantograph is a set of tracing devices similar to facebows which are attached to the mandible and maxilla which records mandibular movements in three planes. A simplified version of such a device is the Denar Cadiax (Whip Mix Co.) (Figure 6.30). The Denar Cadiax consists of a maxillary bow which is strapped to the head. The nasal support is adjusted until the condylar component of the bow sits at the terminal hinge axis position (Figure 6.30, left). The mandibular bow is then attached to the mandibular clutch which is essentially an impression tray, loaded with impression material to keep it securely in place, and an anterior rod to which the bow is attached. A sensor plate is then attached to the maxillary bow and a stylus to the mandibular bow (Figure 6.30, right). Left and right lateral excursive movements are then made by the patient and the Denar Cadiax records these movements both graphically and numerically (Figure 6.31). These readings can then be set on the articulator.

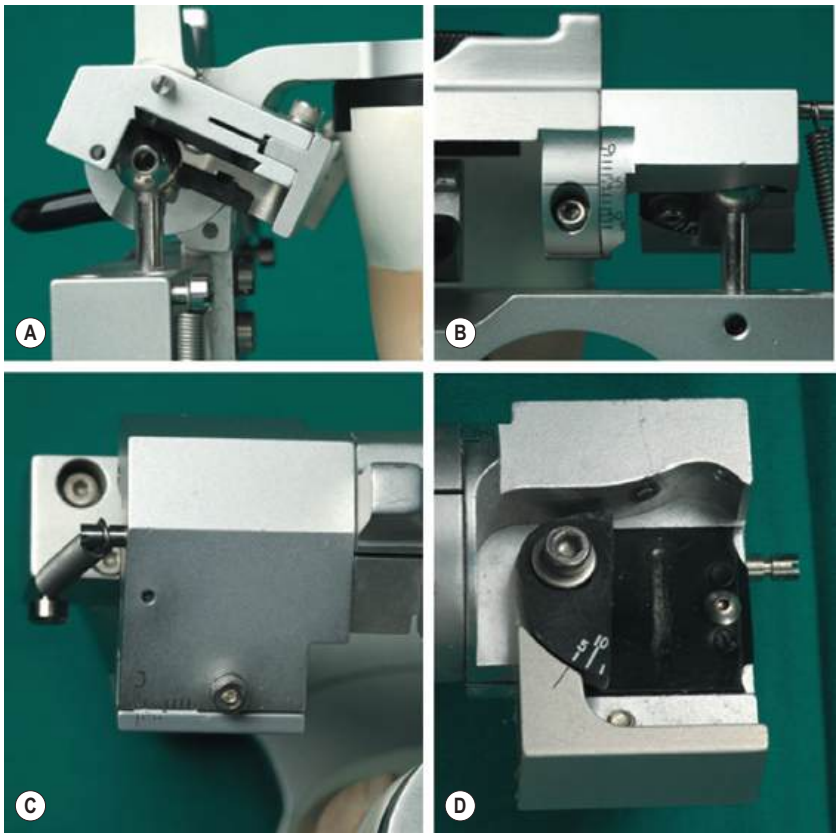


Figure 6.29 Adjustable components on the Denar MK II articulator. Top shows how condylar inclination is adjusted and below how Bennett movement (left) and Bennett angle (right) are adjusted.



Figure 6.30 The Denar Cadiax consists of a maxillary component strapped to the head and a nasal support adjusted until the condylar component of the bow sits at the terminal hinge axis position. The mandibular component is attached to a clutch which is placed over the teeth with a loaded impression tray.



Figure 6.31 The Denar Cadiax gives a graphic recording of the condyle movements seen here for the left condyle. Progressive side shift (Bennett angle) and condylar inclination are illustrated. Numerical values for these angles and immediate side shift (Bennett movement) are also given (inset).

SUMMARY

This chapter has addressed why occlusion is important in restorative dentistry and has attempted to diffuse some of the mystique that has surrounded the subject. Examination of a patient's occlusion is an essential part of any clinical examination and the aspects that should be looked for and recorded in the patient's clinical notes are described, together with the relevance to making restorations whether or not they are direct or indirect. The use of articulators plays an important role in this examination, particularly when the restorative procedures become more complex. Their use is also important for the construction of indirect restorations and, as such, their use has been described in detail. When indirect restorations are fitted the occlusion will need to be checked and adjusted. How this is done is described in Chapter 9. There are also special considerations to managing the occlusion – for example, in reproducing guidance, management of short clinical crowns and reorganizing the occlusion. These are addressed in Chapters 14 and 16.

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